

"Osprey 3.5 V6: High-pressure fuel pump wear sets low fuel rail pressure P0087"

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Area: bulletins

1 Condition

Some Osprey 3.5 V6 vehicles may exhibit extended cranking, a stumble under hard acceleration, and an illuminated malfunction indicator. The PCM will store DTC P0087 "Fuel Rail/System Pressure - Too Low" when commanded rail pressure cannot be maintained under demand. Customers most often report the concern during wide-open-throttle or trailer-towing conditions when fuel demand is highest. Refer to Osprey Fuel Service Manual :: Fuel Pressure and Sensor Faults for the conditions under which P0087 sets.

2 Cause

This condition is caused by mechanical wear of the high-pressure GDI fuel pump, part number HPFP-4120, whose internal cam follower and plunger degrade and reduce delivered rail pressure over time. As HPFP-4120 wears, it can no longer meet the PCM pressure target, so the controller sets P0087. Pump construction and operating-pressure details are documented in Osprey Fuel Service Manual :: High-Pressure Fuel Pump (GDI) .

3 Diagnosis

Run PROC-FUEL-DIAG to compare commanded versus actual rail pressure and to verify low-pressure supply before condemning the high-pressure pump. Confirm that the fuel rail pressure sensor FRS-4180 is reporting accurately, since a skewed FRS-4180 signal can mimic P0087 without any actual pressure loss. The complete diagnostic sequence is provided in Osprey Fuel Service Manual :: Fuel Trim and Pressure Diagnosis Procedure and the broader Osprey Fuel Service Manual :: Fuel System Diagnosis section.

4 Repair Procedure

If supply pressure is normal and FRS-4180 reads correctly but actual rail pressure remains below target, replace the high-pressure pump HPFP-4120. Perform PROC-FUEL-PUMP and follow the Osprey Fuel Service Manual :: Fuel Pump Replacement Procedure , observing all fuel-system depressurization and cleanliness requirements. After installing HPFP-4120, prime the system, clear codes, and verify P0087 does not return under load; see Osprey Fuel Service Manual :: Fuel Delivery and the master Osprey Fuel Service Manual for torque and bleed specifications.